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1 Introduction

The Helium–Neon (He–Ne) laser is a gas laser that operates on the basis of population inversion and stimulated emission in atomic Neon. The laser medium consists of a gas mixture of Helium and Neon enclosed in a discharge tube placed inside an optical resonator.

The excitation mechanism relies on an electrical gas discharge. By applying a high voltage, free electrons are accelerated and excite Helium atoms via electron collisions of the second kind. Due to optical selection rules, the excited Helium atoms populate metastable states, which cannot decay radiatively to the ground state. These long-lived states enable efficient energy transfer to Neon atoms through atomic collisions of the second kind. The excitation energies of these Helium metastable states closely match the energies of specific excited Neon states, so that Neon atoms are selectively excited into the 3s and 2s levels.

Laser action occurs when a population inversion is established between the Neon 3s (or 2s) states and lower-lying 2p states. Because the lifetimes of the s-states are significantly longer than those of the p-states, the inversion condition is fulfilled. Transitions from the 3s to the 2p levels give rise to visible laser lines, most notably the red line at 632.8 nm, which exhibits the highest gain in the visible spectrum. Additional transitions in the infrared spectral range originate from the same upper levels and compete with the visible lines.

The laser medium is embedded in an optical resonator formed by two mirrors, which provides optical feedback and allows standing electromagnetic waves (resonator modes) to build up. Due to Doppler-broadening caused by the thermal motion of the Neon atoms and natural line broadening arising from the finite lifetime of excited states, the gain profile of the laser medium exhibits a Voigt shape. Only resonator modes whose frequencies lie within this gain profile and whose gain exceeds the losses can oscillate.

To control the laser emission, wavelength-selective elements can be introduced into the resonator. In this experiment, wavelength selection is achieved using a Littrow prism and a birefringent crystal, which impose wavelength-dependent losses and thus suppress unwanted laser lines. Brewster windows at the ends of the laser tube ensure a well-defined linear polarization and minimize reflection losses.

2 Experimental setup

The experimental setup consists of a Helium–Neon laser tube mounted on an optical rail and integrated into a linear optical resonator. The laser tube contains a low-pressure mixture of Helium and Neon and is operated by a high-voltage power supply providing the electrical discharge required for excitation.

The resonator is formed by two adjustable mirror mounts. One side acts as a high-reflectivity end mirror, while the opposite side can be configured either with a Littrow prism for wavelength selection or with a standard mirror in combination with a birefringent crystal. Brewster windows are attached to both ends of the laser tube to minimize reflection losses and enforce linear polarization of the laser light.

For alignment purposes, a pilot laser is used to define the optical axis of the resonator. An iris is positioned along the beam path to ensure proper alignment. A photodetector is placed near the Brewster window to monitor the intensity of reflected light during the alignment process.

For wavelength selection, either a Littrow prism or a birefringent crystal can be inserted into the resonator. The Littrow prism replaces one of the resonator mirrors and allows wavelength tuning via angular adjustment. Alternatively, the birefringent crystal is inserted into the resonator together with a conventional mirror, enabling wavelength selection through polarization-dependent phase shifts.

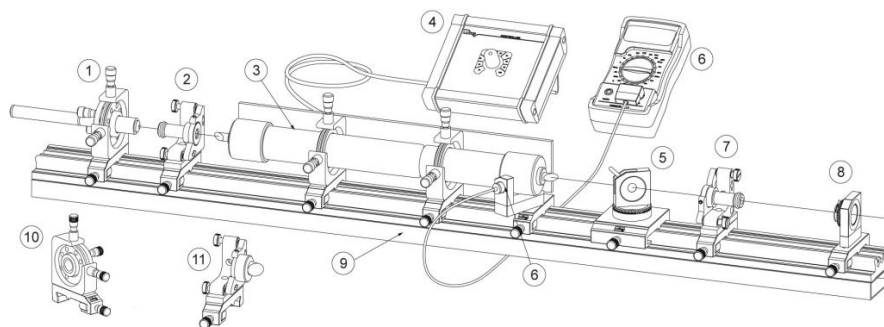


Figure 1: Sketch of the He-Ne-setup (with the birefringent crystal)

3 Setup of the Laser

3.1 Procedure

The laser was aligned following the procedure described in the laboratory manual. First, the optical axis of the resonator was defined using the pilot laser and the iris. The mirrors were adjusted iteratively to ensure that the pilot beam propagated centrally through the laser tube and the Brewster windows.

After completion of the alignment, the pilot laser was turned off and the high-voltage power supply was activated. Although the alignment procedure had been completed, laser oscillation did not start immediately. Therefore, the additional fine-adjustment steps described in the manual were followed until stable laser operation was achieved. After the laser finally started operating, a process called beam-walking was performed in order to maximize the laser output intensity, which was monitored with a laser-power-meter.

After the maximum laser output power had been achieved, the emitted power was measured for all available laser discharge currents. In addition, the emission spectrum of the laser was measured using a spectroscopic software. The spectral data were saved and subsequently plotted for further analysis.



Figure 2: Final alignment of the Helium-Neon-Laser

3.2 Data, results and interpretation

Current [mA]	Laser-Power [μ W]
4.00	111.5
4.25	114.3
4.50	122.7
4.75	123.1
5.00	124.3
5.25	127.3
5.50	128.5
5.75	129.5
6.00	130.1
6.25	130.9
6.50	131.3

Table 1: measured Laser-Power for different currents

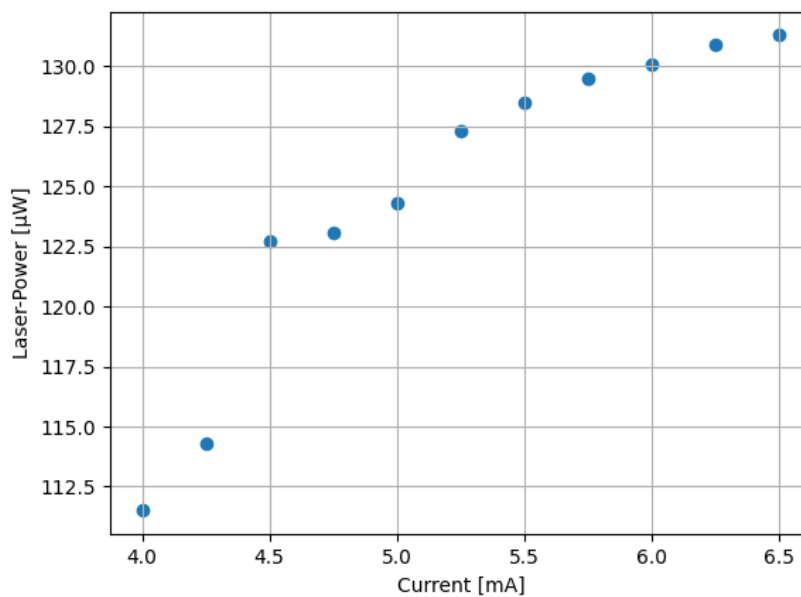


Figure 3: measured Laser-Power plotted for different currents

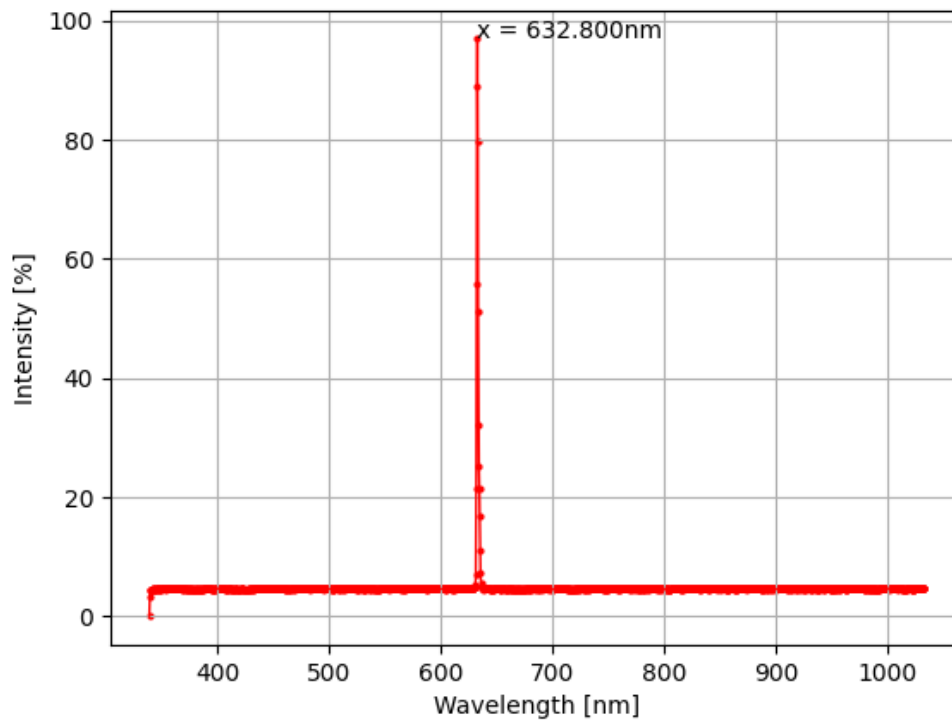


Figure 4: measured intensity for different wavelengths

In the investigated current range, the laser operated well above threshold. The output power increased only weakly with increasing discharge current and showed no further significant increase at higher currents, indicating saturation of the laser gain.

The measured intensity as a function of wavelength exhibits a well-defined, slightly broadened spectral peak with a maximum at 632.8 nm. This value is in excellent agreement with the theoretically expected wavelength of the dominant visible He–Ne laser transition, since this wavelength exhibits the highest gain among all visible laser lines.

4 Wavelength selection with a Littrow prism

4.1 Procedure

For wavelength selection using a Littrow prism, the left resonator mirror was replaced by the prism. After inserting the prism, the laser did not immediately operate stably. Therefore, the alignment and adjustment steps described in the laboratory manual were followed again until laser oscillation was achieved.

Once stable laser operation was obtained, the prism angle was varied while maintaining resonance. During this procedure, beam-walking was once again applied in order to compensate for lateral beam displacement caused by the rotation of the prism and to maximize the laser output intensity.

Despite careful adjustment of the prism angle, no laser emission at wavelengths other than the dominant red line could be observed. This result was expected, since no additional visible laser lines had been detected in the preceding experiments and the red line exhibits the highest gain among the visible transitions of the He–Ne laser.



Figure 5: Operating HeNe-Laser with Littrow prism

4.2 Data, results and interpretation

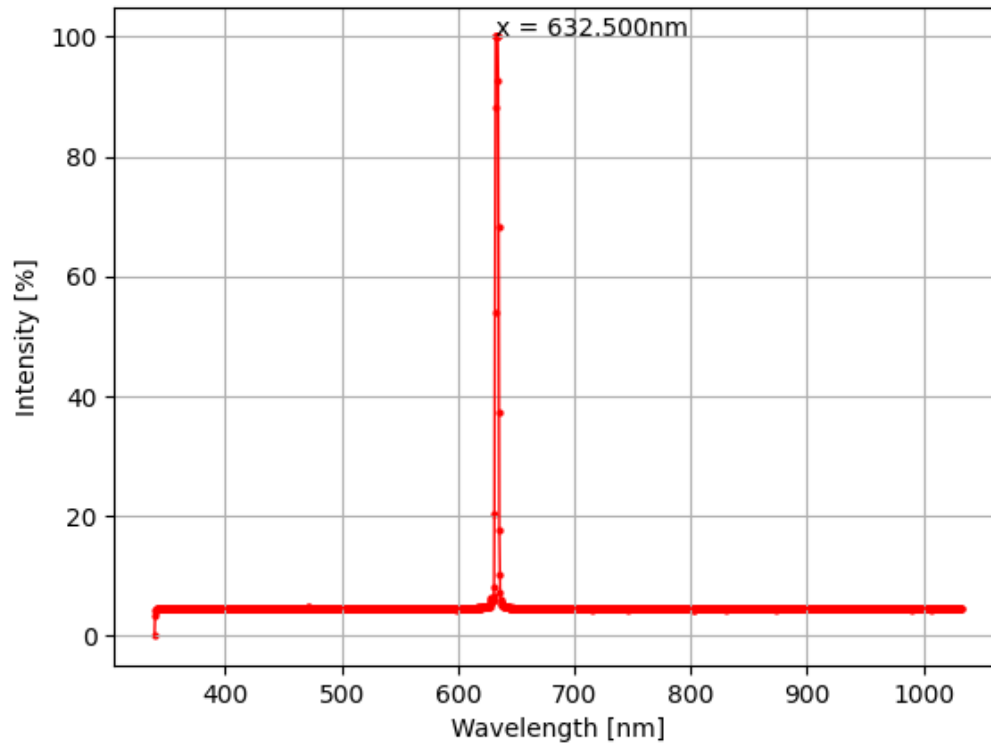


Figure 6: measured intensity for different wavelengths with Littrow prism

In the measured spectrum, only the dominant red laser line at 632.5 nm was observed, while no additional visible laser lines could be detected. This observation is consistent with the visual appearance of the laser during the experiment. The absence of further laser lines is not unexpected and may be related to practical limitations of the experimental setup, such as the condition or age of the laser tube.

5 Wavelength selection with a birefringent crystal

5.1 Procedure

For wavelength selection using a birefringent crystal, the Littrow prism was removed and replaced by a conventional resonator mirror, while the birefringent crystal was inserted into the resonator. The crystal was mounted at the Brewster angle (approximately 57°) in order to minimize reflection losses and to ensure a well-defined linear polarization inside the resonator.

After modifying the setup, the laser did not immediately operate stably. Therefore, the alignment procedure described in the laboratory manual was briefly repeated. Compared to the initial laser setup and the Littrow prism configuration, stable laser operation was restored relatively quickly.

Subsequently, the birefringent crystal was rotated around its optical axis. This rotation influences the phase difference between the polarization components and thereby affects the wavelength-dependent losses inside the resonator. During this procedure, the laser output intensity was continuously monitored and adjusted to achieve maximum output power.

The adjustment lever of the birefringent crystal was rotated over its full available range and the repetition of the observed series of red laser modes was recorded. Throughout the entire adjustment range, only the dominant red laser line was observed, while no orange laser emission could be detected. As discussed previously, this result is consistent with the observations made in the preceding experiments and may be attributed to practical limitations of the experimental setup.

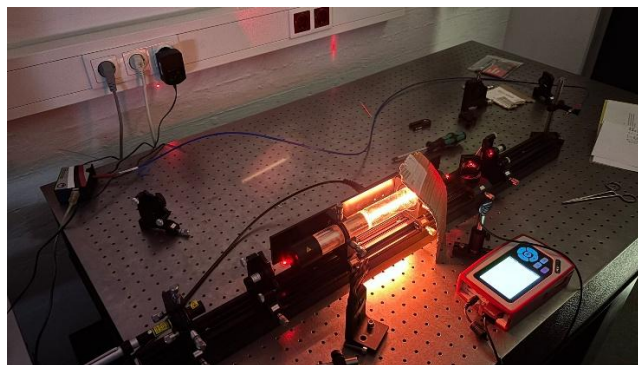


Figure 7: Setup with the Birefringent Crystal at Brewster-Angle

5.2 Data, results and interpretation

Current [mA]	Laser-Power [μ W]
4.00	812
4.25	844
4.50	895
4.75	915
5.00	925
5.25	935
5.50	941
5.75	945
6.00	946
6.25	950
6.50	950
6.75	940

Table 2: measured Laser-Power for different currents with a birefringent crystal

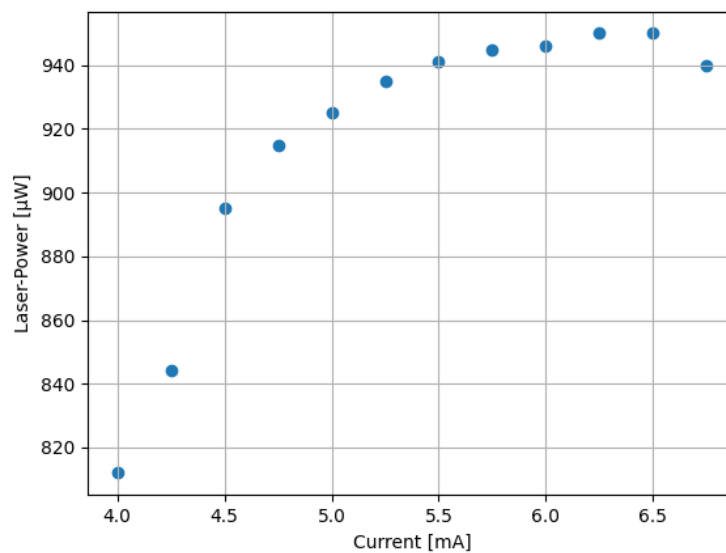


Figure 8: measured Laser-Power plotted for different currents with a birefringent crystal

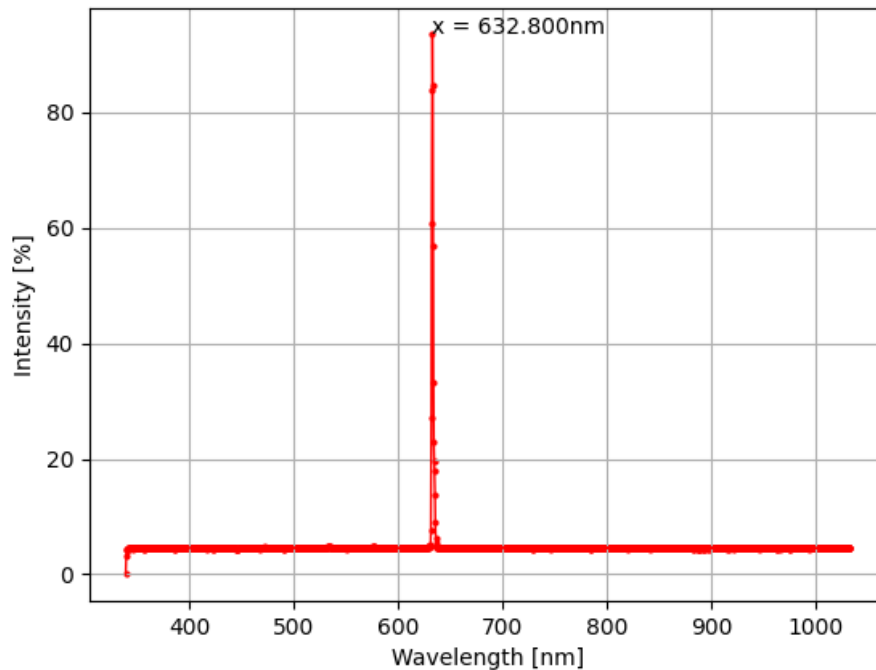


Figure 9: measured intensity for different wavelengths with a birefringent crystal

Once again, the output power increased only weakly with increasing discharge current and showed no further significant increase at higher currents, indicating saturation of the laser gain. The measured intensity as a function of wavelength again shows only a single spectral peak corresponding to the dominant red laser line. No additional visible laser lines were observed. This result is consistent with the visual observations made during the experiment, where only red laser emission was visible throughout the adjustment of the birefringent crystal.

When rotating the birefringent crystal over its full angular range using the adjustment lever, several distinct red laser modes could be observed successively. In total, seven different mode patterns were identified. These modes correspond to different longitudinal resonator modes within the same laser line rather than to different laser wavelengths.

In the following, a theoretical estimation of the expected number of observable modes is performed and compared with the experimentally observed value.

To estimate the number of longitudinal modes oscillating in the He-Ne laser, two quantities are required: the Doppler-broadened gain bandwidth of the Neon transition and the frequency spacing between adjacent longitudinal resonator modes.

The full width at half maximum (FWHM) of the Doppler-broadened gain profile is given by

$$\Delta\nu_{\text{Doppler}} = 2\sqrt{\ln(2)} \frac{v_w}{c} \nu_0,$$

where $\nu_0 = \frac{c}{\lambda_0}$ is the central transition frequency and v_w denotes the most probable thermal velocity of the Neon atoms,

$$v_w = \sqrt{\frac{2kT}{m_{\text{Ne}}}}.$$

The spacing between two adjacent longitudinal resonator modes is determined by the resonator length L and is given by

$$\Delta\nu = \nu(n+1) - \nu(n) = \frac{c}{2L}.$$

For the calculation, the following parameters were used:

- $k = 1.38 \times 10^{-23} \text{ J/K}$
- $T = 293.15 \text{ K}$
- $m_{\text{Ne}} = 3.34988 \times 10^{-26} \text{ kg}$
- $c = 3.0 \times 10^8 \text{ m/s}$
- $\lambda_0 = 632.8 \text{ nm}$
- $L = 0.6 \text{ m}$

Using these values, the ratio between the mode spacing and the Doppler-broadened gain bandwidth yields

$$\frac{\Delta\nu_{\text{Doppler}}}{\Delta\nu} \approx 5,$$

indicating that approximately five longitudinal modes can oscillate simultaneously within the gain profile of the He–Ne laser.

Experimentally, a total of seven distinct longitudinal modes were observed while rotating the birefringent crystal. This result is in good agreement with the theoretical estimate, as both values are of the same order of magnitude. The observed deviation may be attributed to uncertainties in the effective resonator length, approximations in the Doppler broadening model, or additional broadening mechanisms not considered in the simplified calculation, such as pressure broadening or imperfections of the resonator optics.

6 Appendix

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